

Requirements Coverage Report — COMP3211 PIM

Group: 89; member names and student IDs await human input.

Baseline: frozen product source and tests, 4d3508db0447a3ac6346a2a2789da89208f88fd5 (2026-09-30). **Scope:** all eleven Appendix B user stories, FR-01–FR-50 and NFR-01–NFR-05 in docs/srs/SRS.md. The course requires this report in the submitted source-code root; WP08 must copy the final PDF there. This Markdown is the maintained source.

The status **Implemented** means the frozen source contains the behavior and the listed test or WP04 manual session exercises its main path. **Implemented; evidence gap** means source behavior is present but the exact SRS check has no dedicated observed verification. **Partially verified** is used for NFR-02 because the Python 3.11 target was unavailable. Test method names below are exact methods in the frozen test suite; class names locate them: ManagerTests = tests/model/test_manager.py, RecordTests = tests/model/test_records.py, SearchTests = tests/model/test_search.py, SearchIntegrationTests = tests/model/test_search_integration.py, CLITests = tests/controller/test_cli.py, PersistenceTests = tests/storage/test_pim_file.py, and AcceptanceTests = tests/test_acceptance.py. “WP04 session” is the recorded two-process transcript in docs/reports/integration-test-log.md. Evidence for a general path does not imply every suggested edge case was tested.

1. Appendix B user stories

Story	Status	Frozen source file and symbol	Test or manual evidence	Notes
US1	Implemented	model/manager.py::PIMManager._create, .list_all; controller/cli.py::CommandController._dispatch	ManagerTests.test_create_all_types_and_monotonic_ids; AcceptanceTests.test_all_stories_across_save_and_restart	One collection holds all four types.
US2	Implemented	model/records.py::Note; model/manager.py::PIMManager.create_note	RecordTests.test_four_types_trim_values_and_keep_mobile_text; AcceptanceTests.test_all_stories_across_save_and_restart	Note text.
US3	Implemented	model/records.py::Task; model/manager.py::PIMManager.create_task	ManagerTests.test_create_all_types_and_monotonic_ids; WP04 session	Description and deadline.
US4	Implemented	model/records.py::Event; model/manager.py::PIMManager.create_event	ManagerTests.test_create_all_types_and_monotonic_ids; WP04 session	Description, start, alarm.
US5	Implemented	model/records.py::Contact; model/manager.py::PIMManager.create_contact	RecordTests.test_four_types_trim_values_and_keep_mobile_text; WP04 session	Name, address, mobile number.
US6	Implemented	model/manager.py::PIMManager.update; controller/cli.py::CommandController._dispatch	ManagerTests.test_get_list_update_delete_and_invalid_ids; SearchIntegrationTests.test_search_orders_records_and_does_not_mutate; WP04 session	One-field update.

Story	Status	Frozen source file and symbol	Test or manual evidence	Notes
US7	Implemented	model/search.py::parse_criterion, Criterion.matches; model/manager.py::PIMManager.search	SearchTests.test_type_and_all_text_fields_are_case_insensitive, .test_each_time_field_and_comparison_operator, .test_boolean_precedence_parentheses_negation_and_absent_field; WP04 session	All official atom and connector categories.
US8	Implemented	model/manager.py::PIMManager.get, .list_all; controller/cli.py::_details, CommandController._dispatch	CLITests.test_add_list_show_update_delete_and_help; WP04 session	One/all detail output.
US9	Implemented	model/manager.py::PIMManager.delete; controller/cli.py::CommandController._dispatch	ManagerTests.test_get_list_update_delete_and_invalid_ids; WP04 session	Specified ID only.
US10	Implemented	storage/pim_file.py::save_pim; controller/cli.py::CommandController._dispatch	PersistenceTests.test_round_trip_four_types_and_preserve_higher_counter; WP04 session	.pim save.
US11	Implemented	storage/pim_file.py::load_pim; model/manager.py::PIMManager.from_rows	PersistenceTests.test_round_trip_four_types_and_preserve_higher_counter; WP04 session	.pim load after restart.

2. SRS functional requirements

Source paths in this table are relative to `src/`. A symbol after `::` is the exact frozen class, function, method, or module constant. Rows marked with an evidence gap still have an implementation location; the note describes the missing direct check.

ID	Status	Frozen source file and symbol	Test or manual evidence	Notes
FR-01	Implemented	model/manager.py::PIMManager._create, .list_all	ManagerTests.test_create_all_types_and_monotonic_ids	Four types coexist.
FR-02	Implemented	model/records.py::Note; model/manager.py::PIMManager.create_note	RecordTests.test_four_types_trim_values_and_keep_mobile_text	Required text.
FR-03	Implemented	model/records.py::Task; model/manager.py::PIMManager.create_task	ManagerTests.test_create_all_types_and_monotonic_ids; WP04 session	Required description and deadline.
FR-04	Implemented; evidence gap	model/records.py::Event; model/manager.py::PIMManager.create_event	ManagerTests.test_create_all_types_and_monotonic_ids; WP04 session	Both times persist; no dedicated alarm-after-start or past-time assertion. No ordering check exists in source.
FR-05	Implemented	model/records.py::Contact; model/manager.py::PIMManager.create_contact	RecordTests.test_four_types_trim_values_and_keep_mobile_text; WP04 session	Required fields and +00123 preserved.

ID	Status	Frozen source file and symbol	Test or manual evidence	Notes
FR-06	Implemented	model/validation.py::validate_text; model/records.py::Note.__post_init__, Task.__post_init__, Event.__post_init__, Contact.__post_init__	RecordTests.test_four_types_trim_values_and_keep_mobile_text, .test_invalid_direct_construction, .test_unicode_line_separators_are_not_single_line_text; ManagerTests.test_invalid_creation_does_not_consume_id	Shared validator handles all fields; tests sample failure classes across types.
FR-07	Implemented; evidence gap	model/validation.py::parse_local_time, validate_datetime; model/manager.py::PIMManager._create	RecordTests.test_external_time_has_exact_minute_format, .test_invalid_direct_construction	Format and impossible date asserted; past-time acceptance has no dedicated test.
FR-08	Implemented	model/records.py::Contact.__post_init__; model/validation.py::validate_text	RecordTests.test_four_types_trim_values_and_keep_mobile_text; WP04 session	Mobile value is text; no regional pattern is applied.
FR-09	Implemented	model/manager.py::PIMManager._create, .delete, .from_rows	ManagerTests.test_create_all_types_and_monotonic_ids, .test_from_rows_checks_schema_order_and_counters; WP04 session	ID stays above current and loaded counters.
FR-10	Implemented	model/manager.py::PIMManager._create	ManagerTests.test_invalid_creation_does_not_consume_id	Counter advances after validated construction.
FR-11	Implemented	model/manager.py::PIMManager._create	ManagerTests.test_create_all_types_and_monotonic_ids	Repeated x data across types receives distinct IDs.
FR-12	Implemented	controller/cli.py::_arguments, _count, CommandController._dispatch	CLITests.test_add_list_show_update_delete_and_help, .test_quote_escapes_argument_counts_ids_and_eof; AcceptanceTests.test_all_stories_across_save_and_restart	Commands and counts dispatched.
FR-13	Implemented; evidence gap	controller/cli.py::CommandController._dispatch; model/manager.py::_type_name, PIMManager.update	CLITests.test_add_list_show_update_delete_and_help; ManagerTests.test_get_list_update_delete_and_invalid_ids	Mixed-case command/type/field sampled; not every command form repeated in mixed case.
FR-14	Implemented	controller/cli.py::_FIELDS, _count, CommandController._dispatch	CLITests.test_add_list_show_update_delete_and_help, .test_quote_escapes_argument_counts_ids_and_eof; ManagerTests.test_create_all_types_and_monotonic_ids	Schema order and type-specific count.
FR-15	Implemented; evidence gap	controller/cli.py::_arguments	CLITests.test_quote_escapes_argument_counts_ids_and_eof, .test_errors_do_not_mutate_and_next_command_runs	Escapes and bad count sampled; a dedicated unmatched-quote/unsupported-escape CLI assertion is absent.
FR-16	Implemented; evidence gap	model/manager.py::PIMManager.update; controller/cli.py::CommandController._dispatch	ManagerTests.test_get_list_update_delete_and_invalid_ids; SearchIntegrationTests.test_search_orders_records_and_does_not_mutate; WP04 session	One-field replacement proven; no dedicated update-of-every-field/every-type matrix.

ID	Status	Frozen source file and symbol	Test or manual evidence	Notes
FR-17	Implemented	model/manager.py::PIMManager.get, .update	ManagerTests.test_failed_update_is_atomic_and_forbidden_fields	ID/type/absent field rejected without mutation.
FR-18	Implemented	model/manager.py::PIMManager.list_all; controller/cli.py::_summary, CommandController._dispatch	CLITests.test_add_list_show_update_delete_and_help; WP04 session	Ordered, filtered summaries.
FR-19	Implemented; evidence gap	controller/cli.py::_details, CommandController._dispatch	CLITests.test_add_list_show_update_delete_and_help; WP04 session	Detail fields rendered generically; direct show ID assertions do not individually cover all four types.
FR-20	Implemented	model/manager.py::PIMManager.list_all; controller/cli.py::_details, CommandController._dispatch	CLITests.test_add_list_show_update_delete_and_help; WP04 session	Complete details in ID order.
FR-21	Implemented; evidence gap	controller/cli.py::CommandController._print_records	CLITests.test_search_compound_and_no_matches	Source emits No PIRs. for empty list/show; no dedicated assertion for empty list/filter/show-all.
FR-22	Implemented	model/manager.py::PIMManager.delete	ManagerTests.test_get_list_update_delete_and_invalid_ids; WP04 session	Selected ID removed.
FR-23	Implemented	controller/cli.py::_id, CommandController.run; model/manager.py::PIMManager.get	CLITests.test_quote_escapes_argument_counts_ids_and_eof; ManagerTests.test_get_list_update_delete_and_invalid_ids; AcceptanceTests.test_very_long_id_is_reported_and_session_continues	Invalid/absent IDs recover.
FR-24	Implemented	controller/cli.py::_HELP, CommandController._dispatch	CLITests.test_add_list_show_update_delete_and_help; clean-checkout help smoke run	Help lists forms and grammar.
FR-25	Implemented	controller/cli.py::CommandController._dispatch, .run	CLITests.test_quote_escapes_argument_counts_ids_and_eof; AcceptanceTests.test_all_stories_across_save_and_restart	Exit and EOF.
FR-26	Implemented	controller/cli.py::CommandController._dispatch; model/search.py::parse_criterion	SearchTests.test_invalid_expression_is_rejected_whole; CLITests.test_search_compound_and_no_matches	Empty and valid search paths.
FR-27	Implemented	model/manager.py::PIMManager.search; model/search.py::Criterion.matches	SearchIntegrationTests.test_search_orders_records_and_does_not_mutate	Searches current collection.
FR-28	Implemented	model/search.py::TypeCriterion, _Parser._atom	SearchTests.test_type_and_all_text_fields_are_case_insensitive, .test_invalid_expression_is_rejected_whole	Canonical types.
FR-29	Implemented	model/search.py::TextCriterion, _Parser._atom	SearchTests.test_type_and_all_text_fields_are_case_insensitive, .test_invalid_expression_is_rejected_whole	Five fields; empty/unquoted rejected.

ID	Status	Frozen source file and symbol	Test or manual evidence	Notes
FR-30	Implemented	model/search.py::TextCriterion.matches	SearchTests.test_type_and_all_text_fields_are_case_insensitive	Case-insensitive substring.
FR-31	Implemented	model/search.py::TimeCriterion, _Parser._atom	SearchTests.test_each_time_field_and_comparison_operator, .test_invalid_expression_is_rejected_whole	Nine field/operator pairs.
FR-32	Implemented; evidence gap	model/search.py::TimeCriterion.matches	SearchTests.test_each_time_field_and_comparison_operator	datetime ordering used; no dedicated day/month-boundary equality matrix.
FR-33	Implemented; evidence gap	model/search.py::AndCriterion, _evaluate	SearchTests.test_boolean_precedence_parentheses_negation_and_absent_field; SearchIntegrationTests.test_search_orders_records_and_does_not_mutate	True/false paths sampled; no explicit four-row truth-table test.
FR-34	Implemented; evidence gap	model/search.py::OrCriterion, _evaluate	SearchTests.test_boolean_precedence_parentheses_negation_and_absent_field; SearchIntegrationTests.test_search_orders_records_and_does_not_mutate	True/false paths sampled; no explicit four-row truth-table test.
FR-35	Implemented	model/search.py::NotCriterion, _evaluate	SearchTests.test_boolean_precedence_parentheses_negation_and_absent_field	Negation and double negation.
FR-36	Implemented	model/search.py::_Parser.parse, _evaluate	SearchTests.test_boolean_precedence_parentheses_negation_and_absent_field, .test_deep_valid_expressions_evaluate_without_recursion	Grouping and precedence.
FR-37	Implemented	model/search.py::TextCriterion.matches, TimeCriterion.matches, _evaluate	SearchTests.test_boolean_precedence_parentheses_negation_and_absent_field	Missing field yields false; negation tested.
FR-38	Implemented	model/search.py::_tokenize, _Parser.parse, _Parser._atom, parse_criterion; controller/cli.py::CommandController.run	SearchTests.test_invalid_expression_is_rejected_whole; CLITests.test_errors_do_not_mutate_and_next_command_runs	Whole expression rejected before result display.
FR-39	Implemented	model/manager.py::PIMManager.search; controller/cli.py::_summary, CommandController._print_records	SearchIntegrationTests.test_search_orders_records_and_does_not_mutate; WP04 session	Ordered summaries.
FR-40	Implemented	controller/cli.py::CommandController._print_records	CLITests.test_search_compound_and_no_matches	No matching PIRs.
FR-41	Implemented	storage/pim_file.py::_path, save_pim; controller/cli.py::CommandController._dispatch	PersistenceTests.test_round_trip_four_types_and_preserve_higher_counter, .test_wrong_extension_and_failed_save_preserve_existing_bytes	Exact lowercase .pim.

ID	Status	Frozen source file and symbol	Test or manual evidence	Notes
FR-42	Implemented	storage/pim_file.py:: <code>row</code> , <code>save_pim</code>	<code>PersistenceTests.test_round_trip_four_types_and_preserve_higher_counter</code>	UTF-8 JSON, version 1, exact fields.
FR-43	Implemented	storage/pim_file.py:: <code>save_pim</code>	<code>PersistenceTests.test_wrong_extension_and_failed_save_preserve_existing_bytes</code>	Temporary sibling and <code>os.replace</code> ; old bytes retained on failure.
FR-44	Implemented	storage/pim_file.py:: <code>path</code> , <code>load_pim</code> ; <code>controller/cli.py::CommandController.dispatch</code>	<code>PersistenceTests.test_round_trip_four_types_and_preserve_higher_counter</code> , <code>.test_reject_missing_invalid_and_duplicate_json</code> ; <code>CLITests.test_save_load_swap_and_failure_preserves_active_manager</code>	Validation before active-manager swap.
FR-45	Implemented; evidence gap	storage/pim_file.py:: <code>unique_object</code> , <code>_reject_constant</code> , <code>load_pim</code> ; <code>model/manager.py::PIMManager.from_rows</code>	<code>PersistenceTests.test_reject_missing_invalid_and_duplicate_json</code> ; <code>ManagerTests.test_from_rows_checks_schema_order_and_counters</code>	Invalid cases sampled; no exhaustive mutation for every missing/extra key and invalid field.
FR-46	Implemented	<code>model/manager.py::PIMManager.from_rows</code> ; <code>controller/cli.py::CommandController.dispatch</code>	<code>PersistenceTests.test_round_trip_four_types_and_preserve_higher_counter</code> ; <code>CLITests.test_save_load_swap_and_failure_preserves_active_manager</code> ; WP04 session	Replace collection, keep higher counter.
FR-47	Implemented	storage/pim_file.py:: <code>load_pim</code> ; <code>controller/cli.py::CommandController.dispatch</code>	<code>CLITests.test_save_load_swap_and_failure_preserves_active_manager</code> ; <code>AcceptanceTests.test_invalid_input_recovers_in_real_process</code>	Failed load leaves active manager.
FR-48	Implemented	<code>controller/cli.py::CommandController.path</code>	<code>CLITests.test_save_load_swap_and_failure_preserves_active_manager</code> ; <code>AcceptanceTests.test_all_stories_across_save_and_restart</code>	Tests set a process working directory and use relative paths.
FR-49	Implemented	<code>controller/cli.py::CommandController.run</code> ; storage/pim_file.py:: <code>PersistenceError</code>	<code>CLITests.test_errors_do_not_mutate_and_next_command_runs</code> , <code>.test_tab_separated_search_and_invalid_path_recover</code> ; <code>AcceptanceTests.test_invalid_input_recovers_in_real_process</code> , <code>.test_very_long_id_is_reported_and_session_continues</code>	Categories and prompt recovery.
FR-50	Implemented; evidence gap	<code>model/manager.py::PIMManager.create</code> , <code>.update</code> , <code>.delete</code> ; storage/pim_file.py:: <code>save_pim</code> , <code>load_pim</code> ; <code>controller/cli.py::CommandController.dispatch</code>	<code>ManagerTests.test_invalid_creation_does_not_consume_id</code> , <code>.test_failed_update_is_atomic_and_forbidden_fields</code> ; <code>CLITests.test_errors_do_not_mutate_and_next_command_runs</code> , <code>.test_save_load_swap_and_failure_preserves_active_manager</code>	Failure atomicity sampled; no one test spans every failure category.

3. SRS non-functional requirements

ID	Status	Frozen source file and symbol	Test or manual evidence	Notes
NFR-01	Implemented; evidence gap	main.py::main; controller/cli.py::CommandController.run	AcceptanceTests.test_all_stories_across_save_and_restart; WP04 session	Local CLI runs; a network-disabled/no-graphical-session environment was not separately recorded.
NFR-02	Partially verified	main.py::main; model/manager.py::PIMManager; controller/cli.py::CommandController; storage/pim_file.py::load_pim	WP04 full suite: 33 passing on Python 3.13.5 and 3.12.3; docs/reports/code-freeze.md	Version compatibility is a whole-program property. Python 3.11 target unverified: py -3.11 --version found no installed runtime.
NFR-03	Implemented	main.py::main; model/manager.py::PIMManager; controller/cli.py::CommandController; storage/pim_file.py::save_pim	WP04 AST audit of 40 source imports; docs/reports/integration-test-log.md	Whole-source import audit found only standard-library or local imports.
NFR-04	Implemented	model/records.py::Note, Task, Event, Contact; model/manager.py::PIMManager; model/search.py::parse_criterion	ManagerTests.test_create_all_types_and_monotonic_ids; SearchIntegrationTests.test_search_orders_records_and_does_not_mutate; WP04 import audit	Model behavior resides in separate model package.
NFR-05	Implemented	controller/cli.py::CommandController; storage/pim_file.py::save_pim, load_pim	CLITests.test_add_list_show_update_delete_and_help; PersistenceTests.test_round_trip_four_types_and_preserve_higher_counter; docs/design/Design_Document.md	Distinct command and file components.

4. Evidence and remaining gaps

The WP04 suite reported 33 tests, zero failures, and zero errors on Python 3.13.5 and 3.12.3. The real-process acceptance test and recorded two-process session cover US1–US11; neither substitutes for every individual SRS edge-case check. The table identifies those direct-test gaps. No FR or US lacks a frozen implementation location. Python 3.11 compatibility and the NFR-01 network-disabled condition remain unverified; they must not be reported as passed until observed. The separate Test Coverage Report gives the measured model line coverage and its scope.